Fall 2023 / EC571

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Mid-Term Exam 2

Date: November 8, 2023

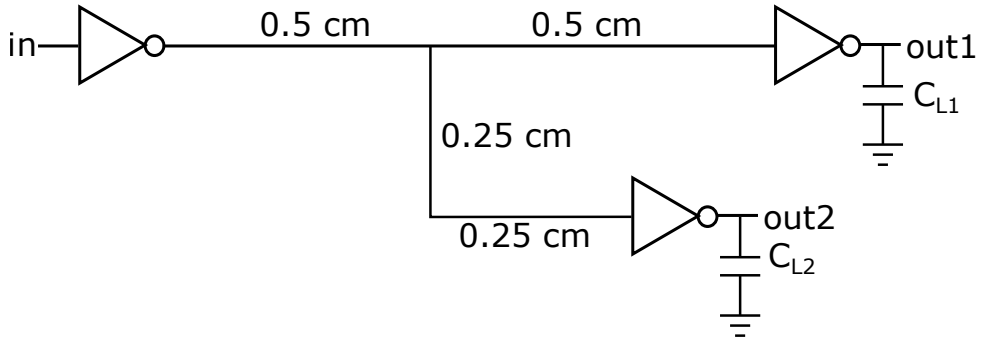
Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
Bonus	5	
Total	100	

1. The duration of the exam is 1 hour 45 minutes.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 6 questions. Please indicate the question number that should not be graded.
5. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
6. Material sharing is not allowed.
7. Calculators are allowed.
8. The exam is closed book close notes, but you are allowed to have two cheat sheets.
9. Boston University Academic Code of Conduct applies.
10. BU ENG is collaborating with several online services, including Chegg, to monitor activity during exams and obtain account information and IP addresses connected to uploaded exam materials and their access. The penalties for cheating are extremely severe including expulsion from BU.

Question 1



A For the circuit shown in the above figure, draw the equivalent RC network that includes all the parts. Assume a lumped model for each interconnect segment. (6 points)

B Write the Elmore delay equations for the entire circuit from in to out1, and from in and out2. (8 points)

C Calculate the Elmore delay for out1 and out2. You can use the following technology parameters for calculating the delay - V_{dd} = 1.5 V, L = 0.25 micron $R_{sq(NMOS)}$ = 10 Kohm/sq, $R_{sq(PMOS)}$ = 20 Kohm/sq, C_G = 1 fF/micron, C_D = 1.5 fF/micron, wire capacitance = 0.25 fF/micron, and wire resistance = 15 ohms/micron. Assume that all inverters use a minimum width of 300 nm for the NMOS. You will need to choose the width of the PMOS transistor such that the low-to-high delay of the inverters matches with the high-to-low delay. Assume a C_{L1} = 10 fF and C_{L2} = 15 fF. (8 points)

Question 2

Consider the circuit shown below. Assume a unit clk-to-q delay for the register, two units of delay for the multiplexer, and two units of delay for each logic block. The registers are positive-edge triggered and have a setup time of one unit. *Hint: Note that we have two parts to the design and you need to consider them separately, but the answer is just based on one of the path.*

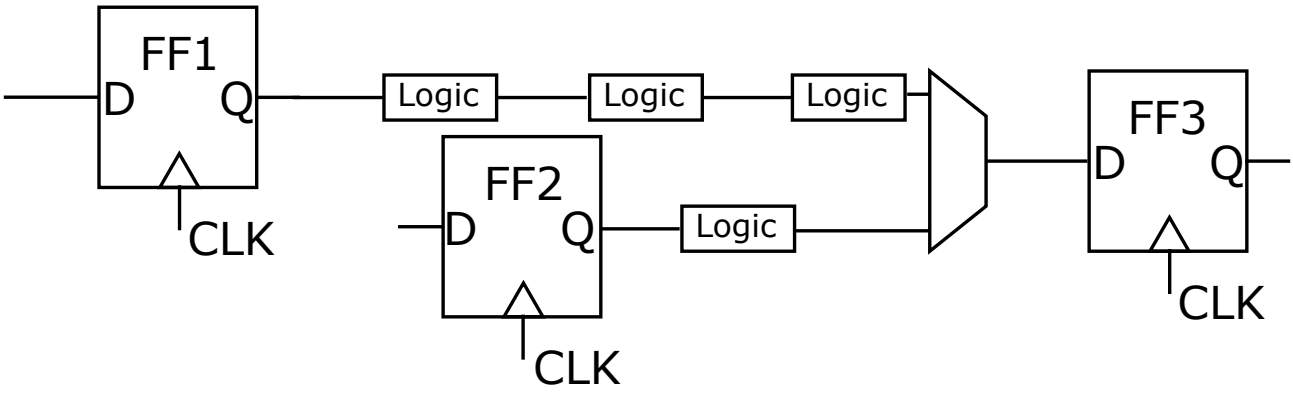


Figure 1

A Determine the minimum clock period for the above circuit. (7 points)

B Assuming a hold time of 2 time unit, what is the maximum positive clock skew that can be tolerated by this circuit? (7 points)

C If the clock input at all FFs has a jitter of ± 1 delay unit, what would be the minimum clock period of the circuit? Assume zero clock skew. (6 points)

Question 3

Sketch the schematics for $f(A,B,C) = \overline{A + B \cdot C}$ using each of the following circuit techniques. You can assume that you have both complemented and non-complemented inputs:

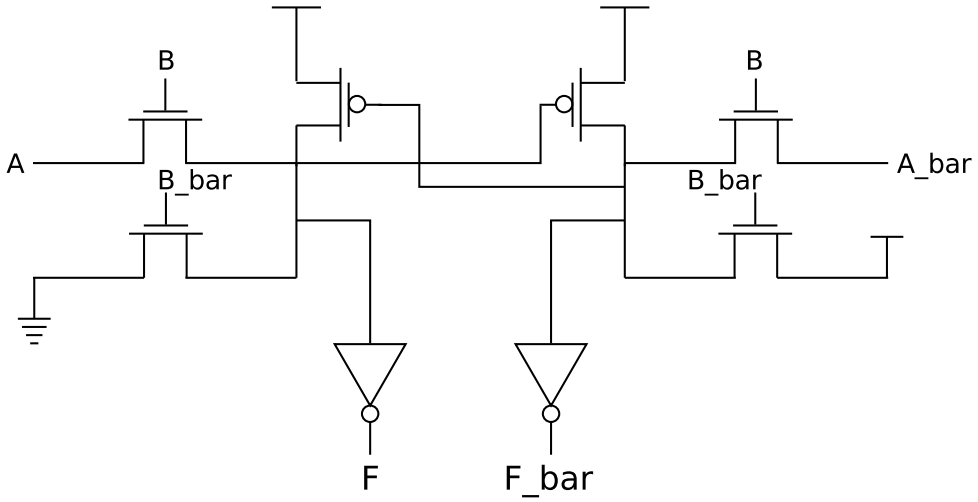
A Static CMOS. (5 points)

B Pseudo-nMOS. (5 points)

C Dynamic. (5 points)

D Pass transistors (including the inverter at the end). (5 points)

Question 4



A Determine the Boolean expression (F(A, B)) for the logic implemented by the complementary pass-transistor logic circuit shown in the figure above. (5 points)

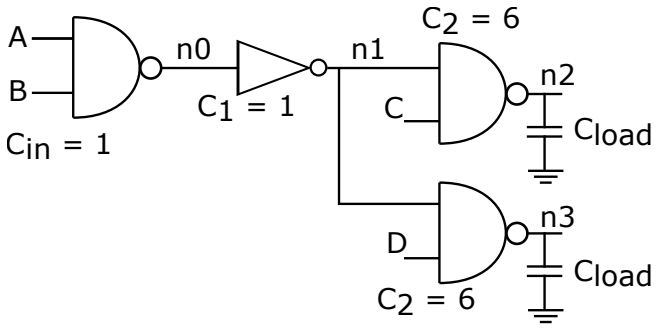
B What is the purpose of the two PMOS transistors in the circuit? (5 points)

C Implement (draw schematic) the same logic (F(A,B,C)) using dynamic logic with PMOS pull-up network. Assume you have both complemented and non-complemented inputs. (5 points)

D Determine the logical effort of the dynamic logic gate that you designed. (5 points)

Question 5

Consider the circuit shown below. Assume $C_D = 0$ throughout this problem.



A Assuming that all of the inputs A, B, C, and D have equal probability of being a 1 or a 0, what are the activity factors (i.e., $\alpha_{0\to1}$) at each of the nodes of the circuit (i.e., n0 – n3)? (6 points)

B Assuming the circuit operates with a supply voltage V_{DD} and a clock frequency f, what is the total dynamic power consumed by this circuit as a function of C_{in}, C_1, C_2, and C_{load} (as labeled above)? Note that you should include the power dissipated by driving the A, B, C, and D inputs. If you aren't sure about your answers for part A), please assume that all of the activity factors are 1/8 (which is not the correct answer to part A). (6 points)

C With the specific sizes shown above, what are the sensitivities S_{C1} and S_{C2}, where $S_x = (\partial Power/\partial x)/(\partial Delay/\partial x)$? Here you can assume that the delay $t_p = t_{inv}(4/3 * C_1/C_{in} + 2 * C_2/C_1 + 4/3 * C_{load}/C_2)$. (8 points)

Question 6

A Figure 2 shows the schematic of a modified Switched Output Differential Structure (SODS) logic. Explain how the logic style operates. Comment on the strengths and weaknesses of this design style as compared to the static complementary CMOS technology in terms of power, performance, area, and reliability. You can assume that both NAND and AND have 2 inputs and you have access to A, B, A' and B' (10 points).

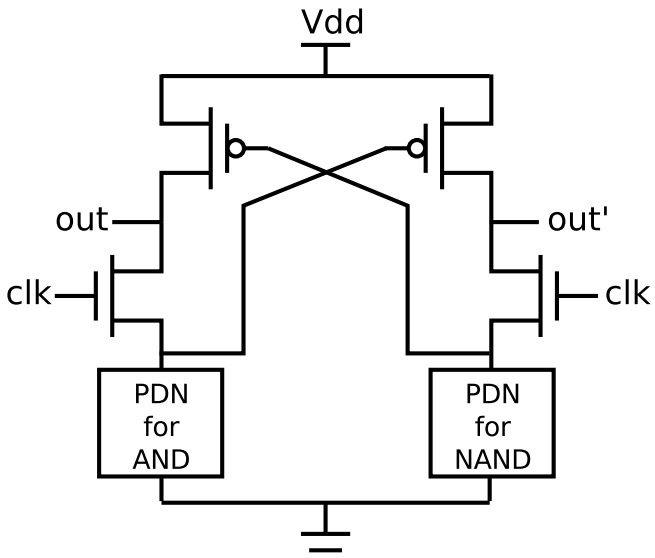


Figure 2

B In your own words, explain the working of a pulse-triggered flip-flop. (5 points)

C Design a negative edge-triggered flip-flop using a positive-level triggered latch and a negative level-triggered latch. (5 points)

Bonus question
Name five cities in the southern hemisphere. (5 points)